

SHWETA PAWAR

✉ nlp.shwet@gmail.com | ☎ +971 58 574 9787 | 📍 UAE (🌐 Open to Relocation)
[LinkedIn](#) | [GitHub](#) | [Medium](#) | [UpWork](#)

PROFILE SUMMARY

Data Scientist | AI/ML Engineer with **8.6+ years of experience** designing and deploying NLP & Generative AI solutions across Finance, Healthcare, Multimedia and Enterprise domains. My specialties are in research and development of **RAG pipelines, LLM fine-tuning, speech tech, and scalable ML systems**. Proven track record delivering **production-ready AI systems and client-focused solutions** on AWS, Azure, and GCP for clients including HSBC, Syngenta, and ADB. Skilled in **MLOps, vector databases, and multi-agent frameworks**. Also a top-rated Upwork freelancer with 100% job success.

TECHNICAL SKILLS

Tools and DB	Python, R, SQL (MySQL, PostgreSQL), Neo4j, GitHub, Streamlit, Spark
GenAI & NLP	RAG, multi-modal RAG, LLMs(GPT, Claude, Llama3, llama3.2-vision, Gemini), VLMs, Transformers Prompt Engineering, Query Rewriting, Guardrails, Knowledge Graphs, Distillation. Conversational AI, Custom NER & Classification, SMLs (GPT, BERT, T5)
GenAI Frameworks	Langchain, Llama-Index, Ragas, Cassandra, PEFT, LoRA, pgvector, MLflow, CrewAI, langgraph, MCP
Speech Technology	Speech Recognition, Text to speech, Custom emotions based TTS, Signal processing.
MLOps & Cloud	FastAPI, Docker, Kubernetes, Databricks, AWS, Azure, GCP
Machine Learning & CV	Statistical Modeling, Supervised & unsupervised Learning, Image Processing, Object Detection
DL & ML Frameworks	PyTorch, Transformers, Hugging face, Tensorflow, OpenCV, sklearn, Pandas, Numpy, Librosa, Matplotlib.

PROFESSIONAL EXPERIENCE

Independent Consultant Lead AI Scientist Jan 2025 - Present	Syngenta GPT - <i>Developed Syngpt, a GenAI platform enabling employees to query HR, IT, Finance, and business units. Built RAG pipeline endpoints on Amazon Bedrock and integrated with agentic AI frameworks (CrewAI, DSPy) using a multi-turn routing system to direct queries across vector DBs. Automated cross-departmental knowledge access, significantly reducing manual support.</i> AltrumAI - <i>Fine-tuned small language model DeBERTa on SageMaker (via Hugging Face Estimator) for Prompt Guard and topic moderation. Deployed scalable endpoints using Amazon Bedrock, Lambda & API Gateway, integrating guardrails into GenAI pipelines. Enabled automated content moderation and prompt filtering in production. Tech Stack: Python, CrewAI, DSPy, Hugging Face, DeBERTa, Amazon Bedrock, SageMaker, Lambda, Prompt Guard.</i> AltaFuturis - <i>Managing pre-sales & client PoCs & MVP, including solution design and documentation. Leading end-to-end engagement with clients and AI engineering teams to deliver enterprise GenAI solutions successfully. Implemented LLMs - Gemini, GPTs, Antropic. Leading end-to-end AI workflow development from data pipelines and model training to scalable deployment (Flask, Docker, AWS) across multiple client projects.</i>
TekSystems (HSBC) Lead Data Scientist Aug 2024 - Jan 2025	KYC Bot Documents Analysis: <i>Developed entity extraction for complex PDFs using RAG and custom rules, to reduce manual review effort. Trained Llama-3.2-Vision for analyzing scanned documents, signatures, and handwritten fields, enabling multimodal extraction and alteration detection. Implemented Chain-of-Thought prompt engineering for business & informal email automation. Built SQL Agents & Decision Agents for structured data extraction and validation. Optimized LLM-based workflows by developing custom rules for enhanced document analysis. Deployed the entire workflow on both GCP and on-premises servers, utilizing FastAPI for efficient</i>

	performance. Collaborated with HSBC stakeholders to align AI-driven KYC automation with compliance and business needs Tech Stack: RAG, LLMs, CoT Prompt Engineering, SQL Agents, Decision Agents, FastAPI, GCP, Document AI, Llama-3.2-Vision.
Aboitiz Data Innovation Sr Data Scientist Oct 2022 - Aug 2024	1. GenAI & LLM based Conversal AI bot for Asian Development Bank: Developed Conversational AI for ADB using GPT, Mistral, Claude, and Llama LLM models. Extracted text/tables from complex PDFs using Texttract and LLMSherpa, generated embeddings with BGE-Large and text-embedding-ada-002. Applied advanced chunking strategies with pgvector indexing to build efficient retrieval pipelines that improved accuracy and minimized hallucinations. Designed evaluation pipelines with MLflow, Ragas, and custom metrics. Improved & optimised the entire pipeline by integrating - query rewriting, meta filtering, reranking, Agents, CoT Prompting and knowledge graphs. Fine-tuned Llama2 LLM model using PEFT, LoRA, QLoRA and Quantization. Deployed full pipeline using Databricks, Spark, FastAPI (async). Tech Stack: GPT, Mistral, Claude, LLaMA, BGE-Large, text-embedding-ada-002, pgvector, LangChain, LlamaIndex, MLflow, Ragas, LLMSherpa, FastAPI, Databricks, Spark, PEFT, LoRA, Quantization, Agents. 2. Explainable and Responsible AI: Designed and implemented an Explainable and Responsible AI framework, leveraging SHAP values for interpretability, error analysis for model improvement, and performance metrics for robust evaluation. Conducted fairness analysis to ensure unbiased outcomes and determined feature importance for enhanced transparency. 3. Smart Procurement Analysis: Developed an intelligent procurement system using SVM, BERT, and FinBERT for text classification. Automated item description categorization and price distribution analysis across Aboitiz business units. Conducted high-level procurement analysis using advanced NLP techniques. Tech Stack: Python, BERT, FinBERT, scikit-learn, Pandas, NumPy
Mango Sciences NLP Data Scientist May 2021 - Oct 2022	NLP-Driven Cancer Characteristic Extraction from Unstructured Data: Developed custom annotation tools for large-scale cancer notes. Built custom NER & regex pipelines to extract lab tests, results, and clinical entities. Trained BERT, Bio-BERT, TF-IDF, Word2Vec, GloVe models for text classification (metastases, ICD10, medication detection). Explored Neo4j knowledge graphs for oncology research. Tech Stack: Python, Pandas, scikit-learn, PyTorch, spaCy, Regex, TF-IDF, Word2Vec, GloVe, Bio-BERT, Neo4j, T5
Upwork NLP & Speech Data Scientist May 2020 - June 2021	1. Smart AI recruitment system tool: Built a profile matching system with skills extraction & categorization (Hard, Soft, Tools). Implemented skills scoring & recommendations using custom NLP pipelines. Deployed as scalable NLP microservices on AWS EC2 with FastAPI. Tech Stack: Python, spaCy, Transformer, skills-ml, FastAPI, AWS EC2 2. Emotion-Based TTS Engine Japanese (Custom Research Implementation) : - Performing signal processing techniques on custom audio-text datasets. - Combined two papers GlowTTS + Global style tokens architecture for custom emotional TTS synthesis. Tech Stack: PyTorch, Tacotron2, GlowTTS, Global Style Tokens, Signal Processing, Librosa, Python, GPU. 3. Implemented Research paper to predict partial UI components using Pytorch: https://arxiv.org/pdf/2001.05308.pdf 4. Document & Legal NLP: Built data pipelines to parse PDFs/Word docs, trained BERT/Word2Vec embeddings with SVM for

	clause classification, and SpaCy & LegalBERT models for entity recognition achieving ~80% manual effort reduction at SR Tekbox. Tech Stack: Python, spaCy, Word2Vec, BERT, LegalBERT, SVM, Pandas, GPT, PyMuPDF.
XECO Media LLP Speech & NLP Data Scientist May 2017 - March 2020	Custom Text-to-Speech Systems: Researched and developed TTS solutions beyond APIs (Google, Microsoft, Amazon). Trained Tacotron, DeepVoice3, WaveGlow, Wavenet, Mellotron models on custom datasets, and built emotion-based TTS (Tacotron2 + Global Style Tokens) deployed on AWS for scalable inference. Education Voice Assistant: Integrated custom ASR & TTS models and developed a RASA-based chatbot; mentored junior team members on dataset preparation. Predictive Analytics & Sentiment Analysis: Built pipelines to scrape competitor data (Twitter, YouTube, News, Blogs) and developed a sentiment engine with 91% accuracy. Designed predictive models that improved hospital readmission accuracy (60% → 95%) and time-series models (ARIMA, Kalman, RNN) for glucose forecasting & early hypoglycemia detection.

EDUCATION

2017	M.Sc. Statistics , Shivaji University, Kolhapur
2015	B.Sc. Statistics , Shivaji University, Kolhapur

CERTIFICATIONS

Multi AI Agent Systems with crewAI	Deeplearning.ai
Knowledge Graphs for RAG	Deeplearning.ai
Natural Language Processing Specialization - 4 Courses	DeepLearning.AI
Neo4j: Graph DB Foundations with Cypher	Udemy
Git Essential Training	LinkedIN
Kubernetes Hands-On - Deploy Microservices to the AWS Cloud	Udemy